

# What a Small Role-Prompted Aggregation Pipeline Does with Instructed Phrases, Planted Figures and Caveats: A Registered Component Study on One 20B Model

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Six registered experiments are in progress (Section 9).

## Abstract

Mixture-of-Agents pipelines send a task to several proposer models and have an aggregator write the final answer. We study a much smaller relative of that design on local hardware: one 20B model writes the answer from three contributions that, in most experiments, come from the same model under one-sentence role prompts. Across 61 numbered experiments registered before they ran, on 18 in-house advisory scenarios, four results hold after correction for small samples. (1) An instruction to label estimates with a named phrase produces the phrase in two thirds of answers (378 runs), and a synonym chosen the day before behaves the same. (2) Two local LLM judges prefer the pipeline’s answer to a single answer from the same model (all-pairs scores 0.61 and 0.58 over 126 pairs judged in both orders) and mark down the instructed phrase (0.41 to 0.45). (3) Given two contributions with different values for a figure, the aggregator states one or the other about evenly, so a second source changes the answer in about half of runs (8 items). (4) A reader model changes its decision when a decisive sentence is appended to the report (11 items). A design assembled from these observations showed no detectable difference in judge preference from the plain pipeline, at about three and a half times the model calls. An earlier draft stated several of these results more strongly. One figure came from a scoring error, several intervals were not computed over scenarios, and the sentence-level judge used throughout agrees little beyond chance (kappa 0.19 to 0.41). No person labelled, judged or read anything. We list the corrections, the limits, and six registered experiments now running.

## 1. Introduction

Pipelines that combine several language-model outputs into one answer are widely used. Mixture-of-Agents (MoA) [1] reports that a stack of six open proposer models and an aggregator outperforms GPT-4o on AlpacaEval 2.0, and attributes the gain to “collaborativeness”: a model shown other models’ drafts writes a better response. In advisory settings (health-care, legal and financial analysis) a common variant gives the proposers specialist roles, on the expectation that specialists contribute knowledge and caution a single generalist would lack.

This project began with one question: does such a council of specialists answer better than one model? At the scale we could run, the answer was no on every measure available to us. On a 60-item battery with checkable answers a single 20B model answered 57 of 59 items correctly and a specialist-plus-aggregator pipeline 56 of 60; two harder batteries stopped at a ceiling (Section 3.6). The work then turned to narrower questions about what the aggregation step does with what it is given:

1. Does an instruction to the aggregator change what it does, or only the words it uses?
2. What do LLM preference judges reward in the aggregated answer?
3. What happens to a figure when one contribution, or two contributions, supply it?
4. How much of the proposers’ qualification reaches the answer?
5. Can a follow-up question to a proposer, or a list of caveats appended to the report, change the outcome?

Prior work has answered parts of these at larger scale. Aggregation pipelines have been evaluated on accuracy as well as preference [1, 2], a single deceptive agent has been shown to damage MoA [3], models are known to adopt supplied content over a correct prior [4, 5], LLM judges are known to penalize expressions of uncertainty [6], and summarization is known to drop scope-limiting detail [8, 9]. What this study adds is small-scale, registered, component-level measurement on one pipeline, together with a record of what went wrong in measuring it.

**What this draft claims.** Four statements, each scoped to one 20B model, 18 in-house scenarios and local LLM judges: an instruction produces the phrase it names (Section 3.1); two local judges prefer the aggregated answer and mark down the

instructed phrase (Section 3.2); with two sources for a figure the aggregator states one of them about evenly (Section 3.3); and a reader model acts on a decisive appended sentence (Section 5.3). We also describe a design assembled from these observations (Section 4) and report that, as a whole, it has been compared with the plain pipeline on judge preference only, with no difference detected (Section 6).

**What changed from v0.2.** An internal review found errors in the earlier draft that the repository itself exposes. They are listed in Appendix A. The three with the largest effect: the reported share of caveats “carried through prose” (0.174) came from a parsing error and is 0.030 under the paper’s own rule; the slope relating caveats in to caveats out was reported with run-level intervals and is close to the ratio of answer length to input length; and the preference shares counted only the minority of pairs each judge decided consistently. The earlier title and abstract described the assembled design as a method that “hardens” MoA. The evidence does not support that description, and it has been removed.

## 2. Setup and Methods

**The system as tested.** The aggregator is gpt-oss:20b, a 20B reasoning model served locally, with the one-sentence prompt “You are the lead analyst. Several specialists have written the contributions below. Write the final answer to the decision-maker’s question.” Three proposers supply contributions. In the early experiments (numbered below 30, scored with a keyword counter that the project later abandoned) the proposers were separate fine-tuned 7 to 8B domain models. In the experiments this paper reports, the proposers are *the aggregator model itself* under prompts of the form “You are a healthcare analyst advising a decision-maker...”, one each for healthcare, legal and finance; in several experiments one stored contribution per role per scenario is reused across runs. Fine-tuned models appear only in the comparison of Section 3.6 and the screening test of Section 5.1. The pipeline is therefore a role-prompted self-ensemble with one aggregation layer. It is not MoA as published, which uses six different proposers of 70B and above over several layers, and results here should not be read as results about that system.

**Comparisons.** The *plain pipeline* is three role-prompted contributions and the aggregator prompt above. The *single answer* is the same model answering the scenario with the same prompt minus the sentence that mentions contributions. The pipeline costs four model calls and the single answer one. No comparison of equal cost (for example several samples of the single model with a vote [15, 14]) was run.

**Scenarios and items.** Eighteen advisory scenarios were written for the project, spanning healthcare, legal and finance content. Experiments use 6 to 18 of them. Three item sets were written for specific tests: 6 planted disagreements, 8 planted figures and 11 decision questions. Scenarios, items, planted facts and the labels the project notebook calls analyst-authored were produced with an AI assistant (see the state-

ment before the references). No public benchmark is used.

**Registration.** Each experiment has a written entry made before it ran, with predictions, the result that would count as failure, and a computation from existing runs of whether the outcome could occur often enough to measure. There are 58 such entries for the 61 numbered experiments. The entries are in the project’s version history ahead of the corresponding results. They are self-registered, and the run records for experiments 25 to 60 were added to the history weeks after their results and carry no timestamps, so the order of registration and data collection cannot be verified from the records. Records made since October 2026 carry time, model digest, seed and settings. Deviations from the registrations are listed in Appendix B.

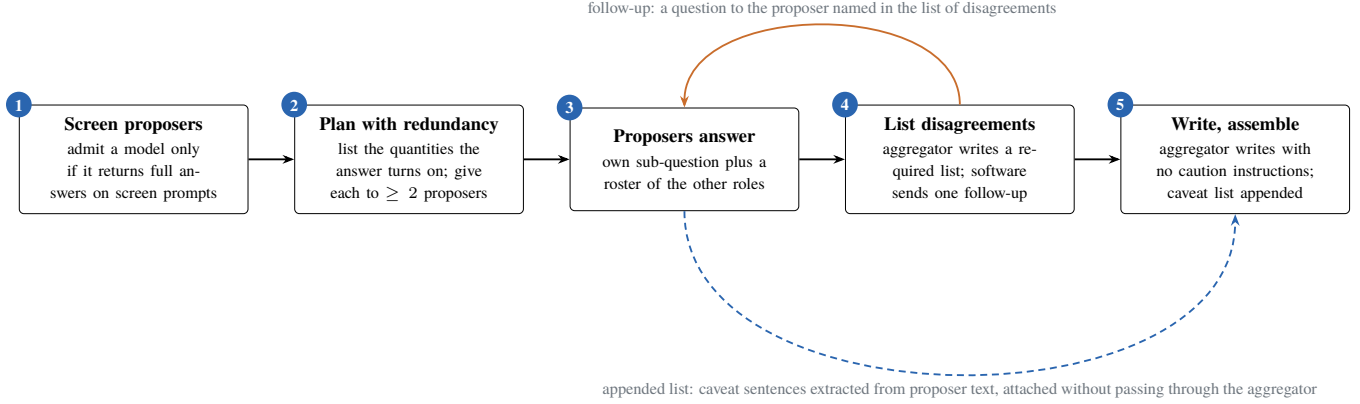
**Instruments.** *Exact phrase:* containment of an instructed phrase. In 2,907 text spans from runs without any instruction, no phrase from the project’s registry of instructed phrases occurred. *Sentence labels:* two judges (gpt-oss:20b and qwen2.5:7b) label each sentence, in batches of ten, for three kinds of qualification (a number labelled as an assumption; different jurisdictions treated separately; a stated condition under which the claim changes), with definitions that name no wording. A sentence counts when both judges flag it. Raw agreement is 0.83 to 0.91, but most sentences carry no qualification: agreement expected by chance is 0.64 to 0.88 and Cohen’s kappa [23] is 0.19 to 0.41 (Appendix D). The labels have not been compared with human labels. Every count that depends on them is provisional and marked †. *Preference:* a judge model sees the question and two answers and names the better one; every pair is judged in both orders. *Exact match* on planted values, after normalisation of number formats. *Reader decision:* a one-word PROCEED or HOLD answer from a reader model.

**Preference score.** For a pair judged in both orders let  $s = 1$  if the first-named side wins both,  $s = 0$  if it loses both and  $s = \frac{1}{2}$  if the judge’s choice follows position. The all-pairs score of a comparison is

$$\text{pref} = \frac{1}{|P|} \sum_{p \in P} s_p, \quad (1)$$

over all pairs  $P$  with two parsed judgments. The judges used here choose the first-listed answer in 0.80 to 0.89 of single judgments, so most pairs score  $\frac{1}{2}$ . The earlier draft reported the share of wins among the pairs with  $s \in \{0, 1\}$  only, which discards 63 to 82% of pairs and overstates differences.

**Statistics.** The unit of analysis is the scenario or the item. For each comparison we average within the unit, take paired differences, and report the mean, a 95%  $t$ -interval and the  $p$ -value of an exact sign-flip test, with the number of units  $k$ . Percentile bootstrap intervals over 6 to 18 clusters, used in the earlier draft, are too narrow [21, 20]. A few intervals are quoted as originally computed and are marked “as first reported”. Table 1 lists the five statements we treat as the paper’s main claims, with Holm-adjusted  $p$ -values [22]; they



**Figure 1:** The design described in Section 4. It keeps the proposer and aggregator structure and adds five parts. The dashed blue lane attaches extracted caveats to the final document; the orange loop sends one follow-up question. Section 5 reports which parts have been tested and how.

were chosen after the results were known, so the adjustment describes rather than protects. Everything else is descriptive.

### 3. What the Plain Pipeline Does

#### 3.1. An Instruction Produces the Phrase It Names

A measurement of an instructed behavior can count compliance with the instruction instead of the behavior. If the instrument detects strings the prompt dictated, the measured quantity  $M$  is the sum of compliance  $C$  and behavior  $B$ ,

$$M = C + B, \quad (2)$$

and separating them needs a design in which the dictated string varies while everything else is held fixed. Three arms were run on 18 scenarios with 7 repeats each (378 runs): the aggregator prompt alone as control, and two arms adding the sentence “When you state a number that is an estimate rather than an established fact, label it using the phrase ‘modeled at’” (form X) or the same sentence with “taken to be” (form Y). Form X had appeared in the project’s prompts for months. Form Y was chosen the day before and appears nowhere in the project’s earlier outputs.

Each instructed arm produced its own phrase in about two thirds of answers:  $+0.667$  [ $+0.450$ ,  $+0.883$ ] for form X and  $+0.698$  [ $+0.504$ ,  $+0.893$ ] for form Y over control, which produced neither (Figure 2, left). Neither arm produced the other’s phrase in 252 runs. A phrase used for months and one chosen the day before behaved alike.

The share of answers containing an assumption-labelled sentence in *other* wording was 0.341 in control, 0.365 under form X (difference  $+0.024$  [ $-0.080$ ,  $+0.128$ ],  $p = 0.75$ ) and 0.317 under form Y ( $-0.024$  [ $-0.131$ ,  $+0.083$ ],  $p = 0.81$ )<sup>†</sup>. Answer length was flat across arms. This second result is weaker than the earlier draft said. It is measured with the two-judge labels, which flagged 73 of 110 sentences containing “modeled at” and 35 of 101 containing “taken to be”. An instrument that misses a third to two thirds of the instructed

phrases could miss an increase expressed in new wording. The supported statement is that the judges saw no change larger than about 0.13; the experiment was registered to detect 0.22 and cannot support a claim of equivalence.

#### 3.2. Local Judges Prefer the Aggregated Answer and Mark Down the Phrase

Three comparisons were judged on the 18 scenarios, 126 pairs each, in both orders. Judge 1 is gpt-oss:20b, which wrote both sides of every pair. Judge 2 is a 30B model from another vendor, chosen from three candidates on whether it returned parseable and consistent judgments, without looking at which side it favored.

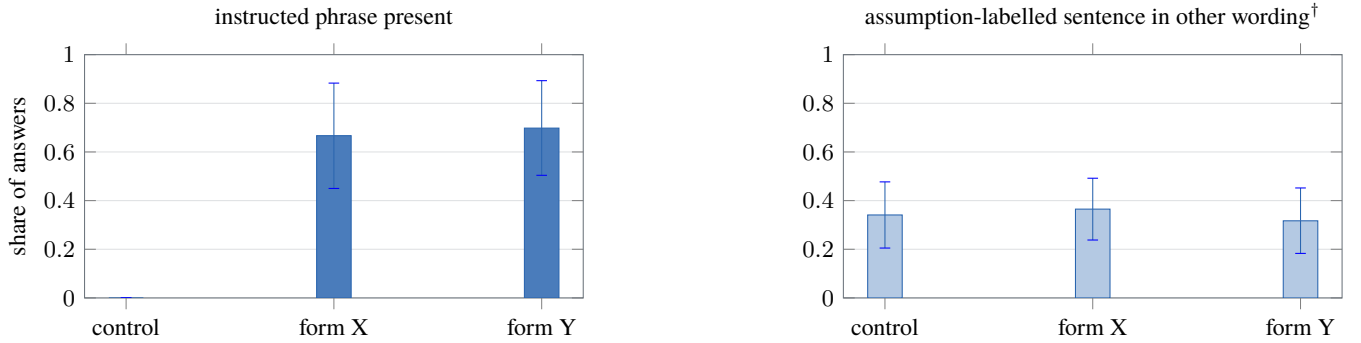
*Pipeline answer against single answer.* Judge 1 gave 36 wins, 8 losses and 82 position-driven splits, an all-pairs score of 0.611 ( $+0.111$  [ $+0.065$ ,  $+0.157$ ] from 0.5,  $p = 0.0005$ ). Judge 2 gave 26, 7 and 93: 0.575 ( $+0.075$  [ $+0.023$ ,  $+0.128$ ],  $p = 0.012$ ). The two sides’ answers differ in length by 1.4%, and restricting to pairs of near-equal length leaves judge 1’s share of consistently decided pairs unchanged (0.815 against 0.818). Which property of the pipeline answer the judges respond to is not identified.

*Instructed phrase against control.* The arms of Section 3.1 differ from control only in the instruction. Judge 1 scored the “modeled at” answers 0.413 ( $-0.087$  [ $-0.144$ ,  $-0.031$ ],  $p = 0.009$ ) and the “taken to be” answers 0.440 ( $-0.058$  [ $-0.118$ ,  $+0.003$ ],  $p = 0.06$ ). Judge 2 scored them 0.448 ( $p = 0.17$ ) and 0.448 ( $-0.052$  [ $-0.092$ ,  $-0.012$ ],  $p = 0.023$ ). Two of the four comparisons reach  $p < 0.05$  and all four lean the same way (Figure 3). The registration had predicted the opposite.

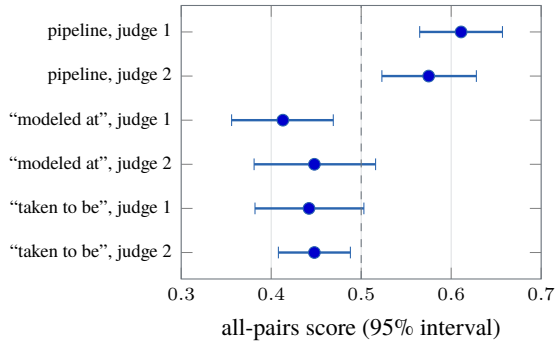
This agrees with larger studies: LLM judges including GPT-4o are biased against epistemic markers [6], and human preference data used in alignment are biased against text that expresses uncertainty [7]. Our contribution is a small replication with a counterbalanced phrase on local judges. Read with Section 3.1, it suggests a tension for systems tuned to-

| Statement                                                                                    | Experiment | Difference [lo, hi]     | $k$          | $p$    | Holm   |
|----------------------------------------------------------------------------------------------|------------|-------------------------|--------------|--------|--------|
| The instructed phrase “modeled at” appears (share of answers, minus control)                 | 41         | +0.667 [+0.450, +0.883] | 18 scenarios | 0.0002 | 0.0012 |
| The primary judge prefers the pipeline answer to a single answer (all-pairs score minus 0.5) | 43         | +0.111 [+0.065, +0.157] | 18 scenarios | 0.0005 | 0.0020 |
| A reader model acts on a decisive appended sentence (flip share, relevant minus irrelevant)  | 51         | +0.691 [+0.427, +0.955] | 11 items     | 0.0039 | 0.0117 |
| The primary judge marks down the answer containing “modeled at” (all-pairs score minus 0.5)  | 43         | −0.087 [−0.144, −0.031] | 18 scenarios | 0.0088 | 0.0176 |
| A second source lowers how often the first source’s figure is stated                         | 47         | +0.450 [+0.095, +0.805] | 8 items      | 0.0312 | 0.0312 |

**Table 1:** The five statements this draft treats as its main claims. Differences are means of scenario- or item-level paired differences with 95%  $t$ -intervals;  $p$  is from an exact sign-flip test; Holm is the adjusted value across the five. The five were selected after the results were known.



**Figure 2:** The phrase-swap experiment (378 runs; 95%  $t$ -intervals over 18 scenarios). **Left:** each instructed arm produces its own phrase in about two thirds of answers; the control arm produces neither, and neither arm produces the other’s phrase (0/252). **Right:** the share of answers with an assumption-labelled sentence in other wording does not differ detectably across arms. This panel is scored with the two-judge sentence labels, which find only 0.66 and 0.35 of sentences containing the two instructed phrases, so a change expressed in new wording could be missed.



**Figure 3:** Preference on all pairs (Equation 1; 126 pairs per comparison, both orders;  $t$ -intervals over 18 scenarios). Judge 1 is gpt-oss:20b, which also wrote every answer; judge 2 is a 30B model from another family. Top two rows: pipeline answer against a single answer. Lower rows: the answer containing the instructed phrase against the control answer. The dashed line is no preference.

ward such judges: the instruction adds a phrase, and the judges prefer the answer without it.

Two limits apply. The pipeline answer costs four calls and the single answer one; sampling one model several times and aggregating is known to help [2, 14], so the role structure has not been shown to add anything. And both judges are local

| Item        | 0   | 1   | 2   | 3   | 4   | 5   | 6   | 7   |
|-------------|-----|-----|-----|-----|-----|-----|-----|-----|
| one source  | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 0.2 |
| two sources | 1.0 | 0.8 | 0.0 | 0.0 | 0.2 | 0.6 | 1.0 | 0.0 |

**Table 2:** Share of five runs in which the planted figure is stated in the answer, per item, with one source and with a second source that gives a different figure (experiment 47).

models whose single judgments mostly follow position; no person judged a pair.

### 3.3. With Two Sources for a Figure, the Aggregator States One of Them

Eight scenarios each received one added figure (a price, a rate, a budget) that appears nowhere in the scenario. One proposer’s text states a value for it, and the question asks for that figure. In the second arm another proposer states a different value. Answers are scored by exact match.

With one source, the aggregator stated the planted value in 36 of 40 runs. With two, it stated the planted value in 18 of 40 and the second value in 20 of 40. At the item level the drop is +0.450 [+0.095, +0.805] ( $p = 0.031$ , 8 items) and the rise in the second value +0.500 [+0.090, +0.910] ( $p = 0.063$ ). The change is all-or-nothing by item (Table 2), and two attempts to predict which source wins, from the model’s own plausibility

| Scenarios, aggregator | in    | out   | length | count |
|-----------------------|-------|-------|--------|-------|
| newer, gpt-oss        | 0.183 | 0.193 | 0.292  | 0.307 |
| newer, phi4           | 0.183 | 0.190 | 0.178  | 0.184 |
| original, both        | 0.133 | 0.103 | 0.300  | 0.231 |

**Table 3:** Share of sentences that carry a caveat in the proposers’ text (in) and in the answer (out)<sup>†</sup>; answer length as a fraction of input length; and caveat sentences out as a fraction of caveat sentences in.

rating and from an elicited map of which role “owns” each quantity, were not evaluable at their sample sizes.

Three points limit the reading. First, repeating the single value is compliance with the question; nothing in the scenario allowed the value to be checked, so the 0.90 rate is not evidence of a failure to verify. That models adopt supplied content readily is established [4, 5]. Second, with two sources the aggregator did not report that two values existed; it stated one. A second source therefore changes the answer in about half of runs without the reader being told there was a choice. Third, when the same design was run with a live planning step and live proposers (experiment 59), the second value was used (2/40 against 19/40, +0.425 [+0.285, +0.565],  $p = 0.008$ ) but the drop in the planted value was not established (25/40 against 18/40, +0.175 [−0.069, +0.419],  $p = 0.22$ ). The earlier draft described this design as “engineered redundancy” that “halves corrupted-value adoption” and “is causal”. Those descriptions are withdrawn. A figure from an early experiment, 0/27 planted values adopted when a clean source remained, is also withdrawn: in that condition 23 of 27 answers stated neither value, and on several items the planted numbers contradicted the scenario text, which the aggregator followed instead.

### 3.4. Caveats In and Caveats Out

Let  $s$  be the number of caveat sentences in the proposers’ text and  $y$  the number in the answer, both counted with the two-judge labels<sup>†</sup>. Within a scenario set they are related by

$$y = w s + c. \quad (3)$$

For gpt-oss:20b the slope  $w$  is 0.172 and 0.326 in the two scenario sets, both with scenario-level intervals above zero (Figure 4). For a second aggregator model, phi4:14b, it is 0.105 and 0.170, and neither interval excludes zero. The earlier draft reported intervals that resampled runs, under which the second model’s slope on the newer scenarios appeared to exclude zero; the claim that the slope “replicates for a second writer” is withdrawn.

The earlier draft also read  $w$  as a loss: “prose transports 0.11 to 0.33 of supplied qualification.” Table 3 shows what the slope mostly reflects. The answer is 18 to 30% as long as its inputs and contains 18 to 31% as many caveat sentences. The share of sentences that carry a caveat is about the same in the answer as in the inputs. A summary is shorter than its sources, and on this evidence caveats are kept roughly in proportion.

Whether a caveat is less likely to be conveyed than an ordinary sentence was not measured; a registered experiment now does that (Section 9). Published work finds that summarization and rewriting do lose scope and certainty information [8, 9, 10], so the question is open here, not answered.

On inputs from which every flagged caveat sentence had been removed and the removal confirmed, the aggregator still wrote at least one caveat sentence in 4 of 12 runs<sup>†</sup>.

### 3.5. A Follow-Up Reply Is Used When It Is Informative

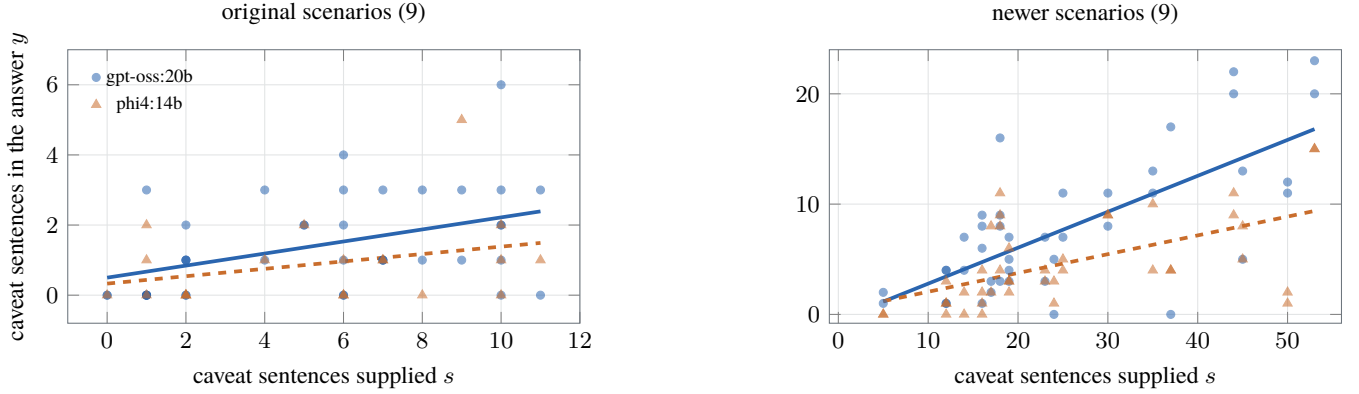
The aggregator was run in two stages: it first wrote a required list of disagreements among the contributions, then the answer. Six items were built in which two proposers take conflicting positions and a fact neither stated settles the question. Between the stages one of three things was added: nothing; a reply in the right style that restates the proposer’s position without the fact; or the same reply with the fact. Two judge models classified which side each answer took (their agreement was 0.70 on the runs analysed and 0.58 overall).

In the runs where the aggregator had named the planted disagreement, it took the fact’s side in 7 of 7 runs with the informative reply, 5 of 15 with no reply and 3 of 15 with the uninformative reply. That contrast conditions on naming, which happened in 7, 15 and 15 of 36 runs although the first stage is identical across arms. Over all 36 runs per arm the counts are 15, 10 and 8. At the item level, informative against none is +0.700 [+0.145, +1.255] among named runs (5 items) and +0.139 [+0.007, +0.271] over all runs (6 items), with exact  $p = 0.125$  in both cases; no item goes the other way. With five or six items no contrast here could be strong. We report it as a consistent direction: when the fact arrives, the aggregator uses it. Naming was the scarce step, at 0.21 to 0.58 of runs across three experiments.

### 3.6. Smaller Observations

**Accuracy.** On a 60-item battery with checkable answers the single model answered 57 of 59 items correctly and a pipeline of the matching fine-tuned specialist plus the aggregator 56 of 60 (paired difference [−0.085, +0.034], as first reported). Two further batteries with script-computed answers (36 multi-step computations, 30 rule-interaction items) put the single model at 0.977 and 0.983, and were stopped before any pipeline run because there was no headroom. We have no evidence that the pipeline gives more correct answers than one model; on the one battery where both ran, the difference was indistinguishable from zero and nominally against the pipeline.

**Fine-tuned proposers.** Two medical fine-tunes of one base model shifted in-domain framework vocabulary by +0.232 [+0.072, +0.412] and +0.109 [−0.056, +0.300] over the base (as first reported). The intervals overlap almost entirely, so the data do not show that the two differ, as the earlier draft claimed. Stronger evidence for the practical point exists: medical adaptations outperform their base models in 12.1% of comparisons and are worse in 38.2% [12].



**Figure 4:** Caveat sentences in the proposers’ text against caveat sentences in the aggregator’s answer<sup>†</sup> (144 runs, 18 scenarios, two aggregator models; each point is one run; lines are least-squares fits within each panel). Slopes with intervals over the nine scenarios of each panel: gpt-oss 0.172 [+0.067, +0.316] and 0.326 [+0.149, +0.477]; phi4 0.105 [−0.013, +0.244] and 0.170 [−0.020, +0.303]. The panels are separate because the two scenario sets supply very different amounts (mean  $s$  of 5.4 against 25.3).

**Self-assessment.** Proposers given a way to defer named the right colleague in 10 of 11 deferrals, and deferred in 0.306 of runs outside their field against 0.083 inside it; the six-item interval for the difference spans zero. A model’s plausibility rating predicted which value it would state at 0.544, recorded as not evaluable, and an elicited map of which role owns each quantity at 0.500 [0.323, 0.739] on 28 runs, recorded as a failed prediction; both tests could detect only a value of 0.68 or more. The earlier draft summarized these as self-assessment that “predicts behavior at chance”. Three underpowered tests do not support that, and the project’s own rule forbids citing an uninformative null as evidence of absence. The question is open; a pre-registered study of small open models finds verbal confidence uninformative for a different reason, saturation at the ceiling [11].

**Agreement between proposers.** The earlier draft said that claims made by two or more proposers reach the answer no more often than claims made by one. That rested on one of four kinds of qualification in an analysis made with the abandoned keyword counter; the other three kinds showed large positive effects of agreement. The claim is withdrawn. Published work finds that models follow the majority of the evidence they are given [5].

**Showing proposers each other’s text.** With stored sibling contributions made visible, off-domain vocabulary moved by  $-0.009$  [−0.048, +0.028] (as first reported, 18 scenarios), against a registered threshold of +0.05. Keeping proposers apart is therefore a cost choice: sibling text adds about 10k input characters per proposer.

#### 4. A Design Assembled from These Observations

The observations above suggest a design with five parts (Figure 1, Figure 5). We describe it as a proposal. Section 5 reports what has been tested, and Table 4 states where the system

that was actually assembled differs from the parts that were tested.

**Screen proposers.** A candidate model is admitted only if every answer on a set of screen prompts reaches 800 visible characters. The earlier draft also listed a probe of domain vocabulary and a format test; neither was validated, and the format test failed to predict anything.

**Assign each key quantity to two proposers.** A planning step lists the quantities the answer turns on and writes each into at least two proposers’ sub-questions. Given Section 3.3, the purpose is to create a visible disagreement when the proposers differ. It does not make the answer immune to a wrong figure, since the aggregator states one of the two about evenly.

**Trigger a follow-up from a required list.** The aggregator writes a list of disagreements before it writes the answer. Software reads the list and sends one follow-up question to the first proposer named. A list the aggregator produces whenever asked (391 of 396 stored runs) is a more reliable trigger than a model’s report of its own uncertainty, whatever that report is worth.

**Append the caveats.** Caveat sentences are taken from the proposers’ text and attached to the final document as a list the aggregator does not rewrite. The aggregator receives no instruction about caution, since Section 3.1 suggests such an instruction adds a phrase and Section 3.2 that judges mark the phrase down.

**Do not tune toward the diagnostics.** Phrase counts, the slope  $w$ , how often disagreements are named, and judge preference on qualified text are recorded and never optimized (Appendix E).

#### 5. Component Tests

Each part was tested in its own experiment with the failure condition written first. Table 5 collects the results. The rows are separate experiments on separate item sets.

| Part          | What was tested                                                                                    | What the assembled system runs                                                                                     |
|---------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| Caveat list   | sentence-level two-judge labels, a median of 11 caveats per document (experiment 48)               | a document-level prompt, at most one quoted sentence per kind per proposer, about 5 per document                   |
| Planning step | told which single quantity to cover (experiment 59)                                                | covers its own list of quantities; coverage checked by a rough word match                                          |
| Follow-up     | item-specific questions; the proposer is handed the deciding fact as private notes (experiment 54) | a generic follow-up for the first listed disagreement that names a role; the proposer holds no private information |
| Screen        | 800-character screen on ten models (experiment 55)                                                 | the same screen on the one model that plays every role                                                             |
| Reply filter  | failed twice                                                                                       | off; replies delivered with a flag                                                                                 |

**Table 4:** The assembled system differs from its tested parts.

| Test (experiment)                                                          | Counts, comparison $\rightarrow$ treatment | Difference [lo, hi]            | $p$   | $k$          |
|----------------------------------------------------------------------------|--------------------------------------------|--------------------------------|-------|--------------|
| Second source, planted: first figure stated (47)                           | 36/40 $\rightarrow$ 18/40                  | +0.450 [+0.095, +0.805] (drop) | 0.031 | 8 items      |
| Second source, planted: second figure stated (47)                          | 0/40 $\rightarrow$ 20/40                   | +0.500 [+0.090, +0.910]        | 0.063 | 8 items      |
| Second source, live chain: first figure stated (59)                        | 25/40 $\rightarrow$ 18/40                  | +0.175 [-0.069, +0.419] (drop) | 0.22  | 8 items      |
| Second source, live chain: second figure stated (59)                       | 2/40 $\rightarrow$ 19/40                   | +0.425 [+0.285, +0.565]        | 0.008 | 8 items      |
| Follow-up, scripted reply, runs that named the disagreement (44)           | 5/15 $\rightarrow$ 7/7                     | +0.700 [+0.145, +1.255]        | 0.125 | 5 items      |
| Follow-up, scripted reply, all runs (44)                                   | 10/36 $\rightarrow$ 15/36                  | +0.139 [+0.007, +0.271]        | 0.125 | 6 items      |
| Follow-up, live proposer, named runs; control stored six days earlier (54) | 5/15 $\rightarrow$ 19/21                   | +0.583 [+0.121, +1.046]        | 0.125 | 5 items      |
| Reader acts on an appended sentence, reader 1 (51)                         | 0.036 $\rightarrow$ 0.727                  | +0.691 [+0.427, +0.955]        | 0.004 | 11 items     |
| Reader acts on an appended sentence, reader 2 (51)                         | 0.091 $\rightarrow$ 0.545                  | +0.455 [+0.104, +0.805]        | 0.063 | 11 items     |
| Document with caveat list against without, judge preference (48)           | 26 wins, 25 losses, 65 splits              | +0.002 [-0.120, +0.124]        | 1.0   | 17 scenarios |
| Screen: the two models that fail are the two with most short outputs (55)  | 2 of 10 fail                               | exact, under random labelling  | 0.022 | 10 models    |
| Assembled system against plain pipeline, preference, judge 1 (61)          | 27 wins, 21 losses, 59 splits              | +0.031 [-0.069, +0.132]        | 0.57  | 18 scenarios |
| Assembled system against plain pipeline, preference, judge 2 (61)          | 5 wins, 10 losses, 93 splits               | -0.023 [-0.060, +0.014]        | 0.31  | 18 scenarios |

**Table 5:** Component tests. Differences are means of item- or scenario-level paired differences with 95%  $t$ -intervals and exact sign-flip  $p$ -values. For preference rows the difference is the all-pairs score minus 0.5.

### 5.1. Screening

Ten local models were screened on two prompts, with the pass or fail recorded before any pipeline run, and each then wrote 18 proposer contributions. The two that failed the screen are the two with the most contributions under 800 characters in the pipeline (17/18 and 6/18; the other eight had 0 or 1 of 18). Under random labelling of two of ten models that has probability  $1/45 = 0.022$ . The earlier draft reported a pooled contrast with an interval that resampled six scenarios; the unit that matters is the model. The screen and the outcome are the same property, short output, measured on different prompts, so this shows that models which answer very briefly on some prompts do so on others. One model that had passed the screen weeks earlier failed it when repeated, so a stored pass should not be reused.

### 5.2. Planning Step

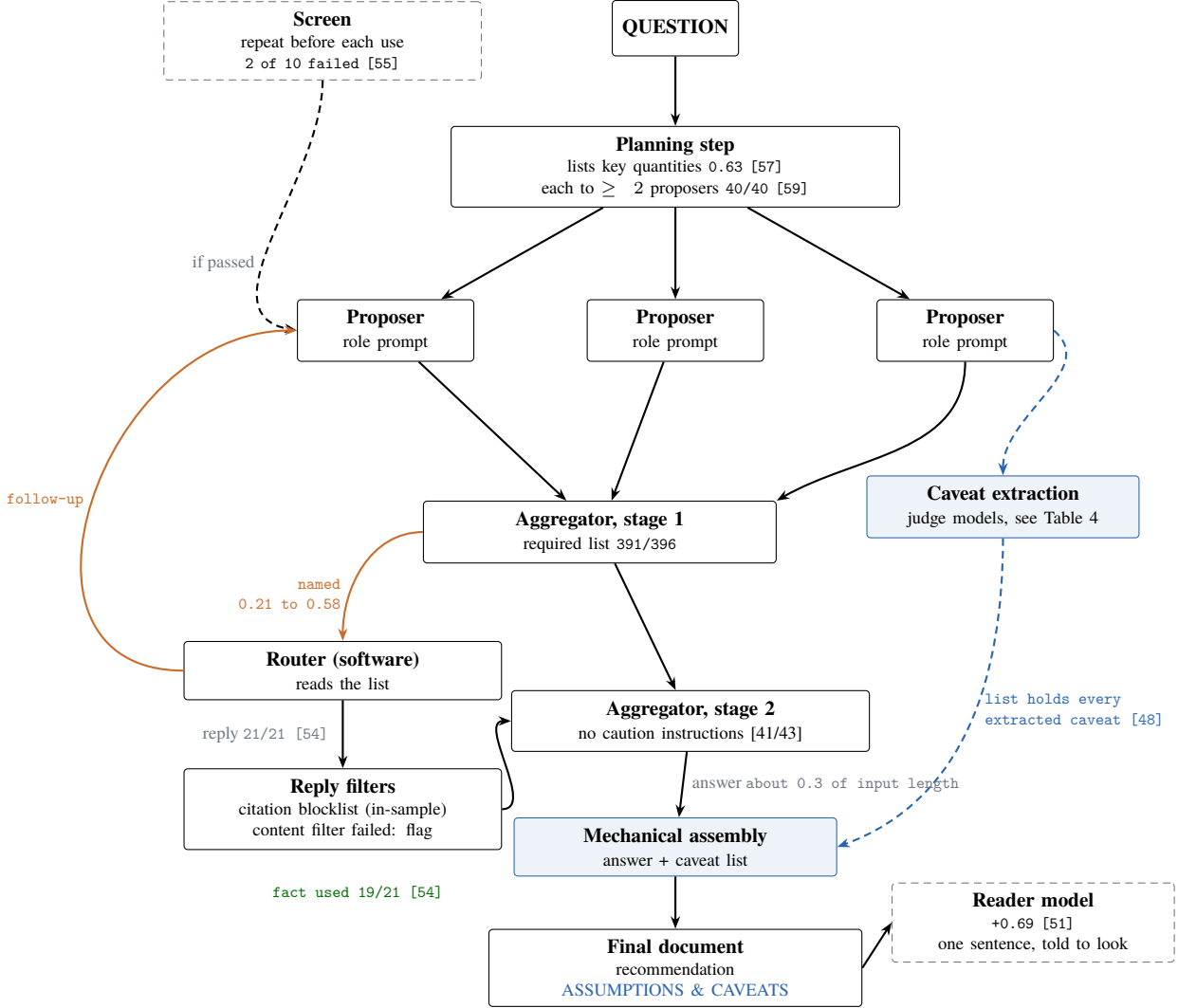
Against 41 labels of key quantities written with the AI assistant, the planning step recalled 0.633 [0.506, 0.758] (as first reported), and did better on quantities that were missing from the scenario (0.813) than on stated ones (0.531). Told which quantity mattered, it wrote that quantity into at least two sub-questions in 40 of 40 plans, and proposers passed privately

supplied values into their contributions in 0.85 to 0.95 of runs. A first attempt stopped at its registered checkpoint with a useful finding: a proposer conveys a privately supplied fact only when its sub-question concerns that fact (2 of 8 otherwise). The downstream result is the live-chain rows of Table 5.

### 5.3. Appended Caveat List

**What the list holds.** The list is built from the proposer sentences that both judges flag<sup>†</sup>, so it contains all of them by construction; the earlier draft reported this “carriage of 1.000” as a result. Across 119 documents, the aggregator’s own prose contains 61 of the 2,037 listed sentence instances word for word (0.030). The earlier draft’s 0.174 came from a parsing error: caveats were read back from the list one line at a time, and the quarter of them that span several lines were cut to a first line that is often a single table character. The corrected figure is a verbatim-copy rate. A caveat restated in the aggregator’s words counts as missing, so it does not measure how many caveats reach the reader. No validated measure of that exists yet.

**Judge preference.** With the list attached to otherwise identical documents, judge 1 gave 26 wins, 25 losses and 65 splits. No penalty was detected and its absence cannot be claimed.



**Figure 5:** The design on one question. Labels give the measurement and the experiment number in brackets. Dashed blue: the caveat list, which the aggregator does not rewrite. Orange: the follow-up loop. The measurements are from separate experiments on separate parts (Section 5); they are not outcomes of the assembled system, which has been evaluated on judge preference only.

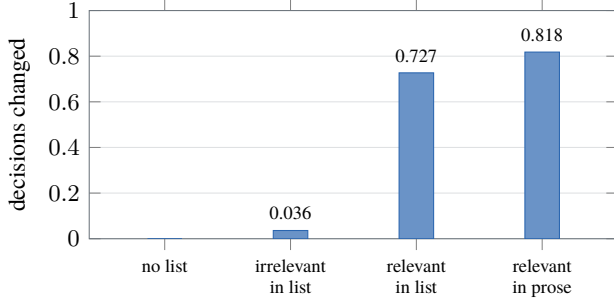
**A reader model uses a decisive sentence.** Eleven decision questions were paired with a sentence that should change the decision and a sentence that should not; some sentences argue for holding and some for proceeding. A reader model answered PROCEED or HOLD. With the relevant sentence as the list, reader 1 (gpt-oss:20b) changed its decision in 0.727 of reads against 0.036 for the irrelevant sentence:  $+0.691$  [ $+0.427, +0.955$ ],  $p = 0.004$  (Figure 6). Placing the same sentence in the prose gave 0.818 (prose minus list  $+0.091$  [ $-0.061, +0.242$ ]). Reader 2 (qwen2.5:7b) gave  $+0.455$  [ $+0.104, +0.805$ ],  $p = 0.063$ .

This shows that a model reader uses appended text. It does not yet show that the design’s list changes a reader’s decision: the reader was the model that wrote the report and was instructed to answer “on the basis of the report and anything attached to it”; the list was one sentence placed directly before

the question, where the system’s lists hold 5 to 53; the planted sentence was a decisive fact written for the item, where the system extracts hedges and assumption labels; the control sentence was shorter (165 against 291 characters on average); and seven of eighteen items were removed at the pilot check. No person read anything.

#### 5.4. Follow-Up with a Live Proposer

Section 3.5 used replies we wrote. Two experiments used a live proposer. In the first the deciding fact sat in the proposer’s own first contribution; the aggregator named the disagreement in 0.21 of runs and the experiment stopped at its registered checkpoint. In the second the proposer was given the fact as private notes at reply time. It conveyed the fact in 21 of 21 replies, and the aggregator took the fact’s side in 19 of those 21 runs (23 of 36 runs overall). The earlier draft compared



**Figure 6:** Share of a reader model’s one-word decisions changed by a planted decision-relevant sentence, by where it is placed (reader 1, 11 items, 55 reads per bar). The list in this experiment is one sentence long and the reader is told to use “anything attached”.

this with the no-reply arm of the scripted experiment, stored six days earlier (5/15; 10/36). The aggregator named the disagreement in 0.583 of the new runs against 0.417 of the old ones, so the batches differ, and the item-level contrast has exact  $p = 0.125$ . We report the counts and claim no effect size. The proposer was handed the fact, so the result shows that the route works when the fact exists in the proposer’s notes.

In 4 of the 21 live replies the proposer added invented legal authority to the correct fact, and the aggregator used the reply anyway. A follow-up experiment found invented citations in 0.100 and 0.150 of replies (120 replies; all on one item, where the same two invented decisions recurred). A blocklist of those names caught every case in 141 stored replies, which shows only that the filter runs as written, since its names were taken from those replies.

In 54 runs of the assembled system on the scenarios themselves, with no planted disagreement, the list named a role and a follow-up went out in 46 (0.85).

### 5.5. Parts That Failed or Were Dropped

A filter meant to drop replies that add nothing failed its registered criteria twice: a two-judge version dropped 3 of 6 uninformative replies, and a stricter version dropped all six but also rejected 52% of genuine replies. Replies are delivered with a flag instead. The ownership map and the plausibility rating (Section 3.6) were not adopted. The rule that proposers do not see each other’s text lost its original rationale (Section 3.6) and stands on cost.

## 6. The Assembled System Against the Plain Pipeline

The assembled system was run three times on each of the 18 scenarios and its documents were paired with stored answers from the plain pipeline (108 pairs, both orders, two judges). On all pairs judge 1 scored the assembled system 0.528 (+0.031 [−0.069, +0.132],  $p = 0.57$ ) and judge 2 scored it 0.477 (−0.023 [−0.060, +0.014],  $p = 0.31$ ). No difference was detected.

Four things limit this result. No equivalence margin was

registered, so it does not show that the two are equally preferred. Judge 2 decided only 15 pairs consistently (5 for the assembled system, 10 against), in 9 of 18 scenarios, and its earlier reported interval of [0.083, 0.562] does not exclude a large loss; the earlier draft’s statement that large effects were excluded in both directions holds for judge 1 only. The plain pipeline’s answers had been stored three weeks earlier. And the comparison is a bundle: several parts change at once.

**Cost.** One question takes about fourteen model calls in the assembled system against four in the plain pipeline, and 14 to 18 minutes on the hardware used. The earlier draft’s phrase “at no cost” is withdrawn.

**What has not been measured.** The assembled system’s effect on wrong figures, on caveats reaching a reader, or on resolving disagreements has not been measured as a system. A registered experiment on wrong figures is in progress (Section 9).

## 7. Limitations

**One model in every role.** Proposers and aggregator are the same 20B model in the experiments that carry the paper’s claims. Results may not hold for different or larger models, or for proposers that differ from the aggregator.

**The baseline is not the published method and is not matched for cost.** See Section 2.

**No human measurement.** Every judge is a local model of 7 to 30B and every reader is a model. No sentence label, preference or decision comes from a person, and the items and several label sets were written with the same AI assistant that designed the experiments, which is the instrument-independence problem of Equation 2 at one remove.

**The sentence instrument is not validated.** Its chance-corrected agreement is low and it misses a third to two thirds of the instructed phrases. Results marked † depend on it.

**Few units.** Claims rest on 5 to 18 items or scenarios, all written in-house. Several effects are all-or-nothing by item.

**Constructed outcomes.** In several component tests the outcome was close to guaranteed by the design: a question that asks for the planted figure; a list scored against the sentences it was built from; a decisive sentence placed before the question; a proposer handed the deciding fact. These tests show that a route works. They do not show how the pipeline behaves on inputs nobody arranged.

**Registration.** The registrations are the author’s own, in the author’s repository, and until October 2026 the run records could not show when they were made. Deviations occurred (Appendix B).

**Many tests.** About ninety registered predictions were made over the program, of which about 35 were recorded as failed. The five claims of Table 1 were selected afterwards.

**Error rate of the record.** The project notebook logs four errors in code that printed verdicts, three text-matching errors, five token-budget failures, one illustrative quotation in an early draft that was not real data, and one experiment run on the

wrong condition. The review behind this revision found two more defects in headline numbers. Others may remain.

## 8. Related Work

**Aggregation pipelines.** MoA [1] reports preference-judged results on AlpacaEval, MT-Bench and FLASK and, in an appendix, accuracy on MATH by layer. Self-MoA [2] shows that aggregating repeated samples of the single best model outperforms mixing different models, on AlpacaEval and on MMLU, CRUX and MATH, and sampling one model many times with a vote scales with the number of samples [14, 15]. These are the missing baselines of Section 3.2. A single strongly prompted agent can match multi-agent discussion [19]. The earlier draft said that every published MoA number came from a preference judge; that was wrong.

**Wrong inputs.** Wolf et al. [3] insert one deceptive agent into a three-layer MoA and measure the damage on AlpacaEval 2.0 (win rate 49.2% to 37.9%) and on QuALITY accuracy, with voting-style defenses that recover most of it. Models adopt incorrect supplied content over a correct prior more than 60% of the time [4] and are highly receptive to coherent external evidence [5]. The earlier draft said no published evaluation measures whether a corrupted proposer corrupts the answer; that was also wrong. Section 3.3 is a small numeric-value variant with a second-source control.

**Judges and uncertainty.** LLM judges including GPT-4o are biased against epistemic markers [6]; human preference data are biased against text that expresses uncertainty [7]; judges show position bias that varies by model [16], length correlations under preference optimization [17], and sycophancy [18]. Our judges’ position bias should be read as a property of the small local judges tested.

**Loss of qualification.** LLM summaries overgeneralize scientific findings in 26 to 73% of cases [8], rewriting raises expressed certainty more often than it lowers it [9], and handoff summaries between agents keep facts while weakening the conditions on their use [10]. Section 3.4 does not yet add a comparable measurement.

**Specialist models and confidence.** Medical adaptations rarely outperform their base models [12]. Verbal confidence in small open models fails a pre-registered validity screen [11].

**Registration.** Pre-registration of experiments with AI agents has a proposed template that lists outcome-contingent redesign among the main risks [13]; the deviations in Appendix B are instances. We are not aware of an earlier experiment that swaps the phrase named by an instruction under otherwise identical prompts, or of one that tests an appended caveat list with a matched irrelevant control.

## 9. Registered Experiments in Progress

Six experiments were registered on 1 October 2026 and are running; a seventh is registered as a design only. Each was redesigned for one person on one machine, and each addresses

a limitation above.

Experiment 62 decides whether the results marked † stand. Experiments 63 and 67 replace the reading of Section 3.3 with a test in which checking is possible and the outcome is a computed number. Experiment 66 supplies the comparison Section 3.4 lacks. Human readers, a second annotator, an independent rerun of the analysis and a much larger model are beyond what one person can do and remain open.

## 10. Conclusion

On one 20B model and eighteen in-house scenarios, four behaviors of a role-prompted aggregation pipeline were observed under registered tests: an instruction to mark estimates produces the phrase it names; two local judges prefer the aggregated answer to a single answer and mark the instructed phrase down; given two sources for a figure, the aggregator states one of them without saying there were two; and a reader model acts on a decisive sentence appended to the report. A design assembled from these observations was indistinguishable from the plain pipeline on a weak preference test and has not been tested on any other outcome. Whether any of it helps a person making a decision is unknown. The measurement lessons may be the more durable result: an instrument that counts strings the prompt dictated measures compliance; raw agreement between judges says little when one answer dominates; intervals over a handful of scenarios need methods built for few clusters; and a number that lands inside the expected range still needs checking.

## Use of AI Assistance

An AI assistant (Claude, Anthropic) was used throughout this project. It proposed and drafted most of the registrations and the experiment code, wrote the scenarios, test items, planted facts and the labels the project notebook calls analyst-authored, ran the analyses, conducted the three internal audits and the October 2026 review that led to this revision, and drafted this text. The author directed the work, decided what was run and what is claimed, and takes responsibility for the content. Because one source wrote the items and labelled several outcomes, results that depend on those labels should be read with that in mind.

## Data and Code

All registrations, run records, analysis code and the project notebook are in the project repository. The corrected figures in this draft are produced by `train/run_stage0_corrections.py` (output in `bench/analysis/stage0/`) at commit `bb7fcab`. The review and its reproduction script are `docs/BOARD_REVIEW_2026-10-01.md` and `bench/analysis/board_review/checks.py`. The experiments of Section 9 are registered at commit `6402fc0`. Neither script calls a model.

| No. | Question                                                                                   | Design in one line                                                                                                                                                              |
|-----|--------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 62  | Do the sentence judges agree with a person?                                                | 404 stored sentences labelled blind by the author twice, a week apart, and by six local judges; bars: retest kappa 0.60, F1 0.80                                                |
| 63  | Does the aggregator catch a wrong figure it could check?                                   | 61 questions with computed answers; a wrong input placed where the case contradicts it, where it is the only source, and beside a right or a second wrong one; scored by script |
| 64  | Do other reader models act on an appended caveat, without the instruction, in a long list? | six reader models, the 11 items of experiment 51, lists of 1, 5 and 10 sentences, a prompt with the attachment clause removed                                                   |
| 65  | Does the judge preference repeat with judges from other families?                          | four more local judges on the 378 stored pairs, all-pairs scoring, judges that only follow position excluded by a rule fixed in advance                                         |
| 66  | Are caveats conveyed less often than ordinary sentences?                                   | 285 caveat sentences and 285 length-matched ordinary sentences, two judges rate whether each is conveyed; checked against 120 pairs labelled blind by the author                |
| 67  | Does a second source help in the assembled chain?                                          | the 61 questions of experiment 63 through the full chain, wrong figure in one proposer’s notes, right figure in a second’s; scored by the computed answer                       |
| 68  | Is the pipeline more accurate than the same model sampled four times?                      | design only: public items screened for headroom, four arms of equal cost                                                                                                        |

**Table 6:** Experiments registered on 1 October 2026. Results will be reported in the next revision whatever their direction.

## Impact Statement

This work concerns language-model pipelines in advisory settings. Its findings are limited to 7 to 30B local models, English advisory scenarios written for the project, and small numbers of items, and they involve no human participants. They should not be used to justify deploying such a pipeline for real advice. The observation that an aggregator given two different figures states one without flagging the conflict is a reason for caution wherever such pipelines are used.

## References

- [1] J. Wang, J. Wang, B. Athiwaratkun, C. Zhang, J. Zou. Mixture-of-Agents enhances large language model capabilities. *arXiv:2406.04692*, 2024.
- [2] W. Li, Y. Lin, M. Xia, C. Jin. Rethinking Mixture-of-Agents: is mixing different large language models beneficial? *arXiv:2502.00674*, 2025.
- [3] L. Wolf, S. Yoon, I. Bogunovic. This is your Doge, if it please you: exploring deception and robustness in mixture of LLMs. *arXiv:2503.05856*, 2025.
- [4] K. Wu, E. Wu, J. Zou. ClashEval: quantifying the tug-of-war between an LLM’s internal prior and external evidence. *arXiv:2404.10198*, 2024.
- [5] J. Xie, K. Zhang, J. Chen, R. Lou, Y. Su. Adaptive chameleon or stubborn sloth: revealing the behavior of large language models in knowledge conflicts. *ICLR*, 2024.
- [6] D. Lee, Y. Hwang, Y. Kim, J. Park, K. Jung. Are LLM-judges robust to expressions of uncertainty? Investigating the effect of epistemic markers on LLM-based evaluation. *NAACL*, 2025.
- [7] K. Zhou, J. D. Hwang, X. Ren, M. Sap. Relying on the unreliable: the impact of language models’ reluctance to express uncertainty. *ACL*, 2024.
- [8] U. Peters, B. Chin-Yee. Generalization bias in large language model summarization of scientific research. *arXiv:2504.00025*, 2025.
- [9] C. G. Belem, S. Wu, H. Yao, M. Steyvers, S. Singh, P. Smyth. From ‘may’ to ‘is’: certainty distortion in language model rewriting. *EMNLP*, 2026.
- [10] Y. Wang, A. Goyal, E. Chandrasekharan, H. Sundaram. Facts without rules: boundary metadata collapse in multi-agent LLM handoffs. *arXiv:2608.29028*, 2026.
- [11] J.-P. Cacioli. Verbal confidence saturation in 3–9B open-weight instruction-tuned LLMs: a pre-registered psychometric validity screen. *arXiv:2604.22215*, 2026.
- [12] D. P. Jeong, S. Garg, Z. C. Lipton, M. Oberst. Medical adaptation of large language and vision-language models: are we making progress? *EMNLP*, 2024.
- [13] M. Vaccaro. Preregistration for experiments with AI agents. *ICML*, 2026.
- [14] J. Li, Q. Zhang, Y. Yu, Q. Fu, D. Ye. More agents is all you need. *Transactions on Machine Learning Research*, 2024.
- [15] X. Wang, J. Wei, D. Schuurmans, Q. Le, E. Chi, S. Narang, A. Chowdhery, D. Zhou. Self-consistency improves chain of thought reasoning in language models. *ICLR*, 2023.
- [16] L. Zheng et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. *NeurIPS Datasets and Benchmarks*, 2023.

- [17] P. Singhal et al. A long way to go: investigating length correlations in RLHF. *arXiv:2310.03716*, 2023.
- [18] M. Sharma et al. Towards understanding sycophancy in language models. *arXiv:2310.13548*, 2023.
- [19] Q. Wang et al. Rethinking the bounds of LLM reasoning: are multi-agent discussions the key? *arXiv:2402.18272*, 2024.
- [20] E. Miller. Adding error bars to evals: a statistical approach to language model evaluations. *arXiv:2411.00640*, 2024.
- [21] A. C. Cameron, J. B. Gelbach, D. L. Miller. Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427, 2008.
- [22] S. Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979.
- [23] J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.

## A. Corrections from Draft v0.2

Each item gives the earlier statement and what the repository shows.

**1. Caveats carried through prose.** Earlier: the appendix “carries 1.000 of proposer caveats against 0.174 through prose.” The 0.174 was a parsing error (Section 5.3); the verbatim rate is  $61/2,037 = 0.030$ , and 1.000 is true by construction.

**2. Intervals.** Earlier: “every confidence interval is a cluster bootstrap over cases.” The slope intervals resampled runs. Over scenarios, two of four exclude zero and the second aggregator model excludes zero in neither scenario set (Figure 4).

**3. Head-to-head.** Earlier: both judges’ shares span 0.5 “with large effects excluded in both directions.” The second judge’s interval was  $[0.083, 0.562]$  from 15 pairs and does not exclude a large loss.

**4. Order of registration and runs.** Earlier: “temporal ordering is verifiable from version history.” Run records for experiments 25 to 60 were first committed on 24 September 2026, weeks after their results, and carry no timestamps.

**5. Prior work.** Earlier: “Every published MoA number is produced by an LLM preference judge” and “No published evaluation measures ... whether a corrupted proposer corrupts the final answer.” Both are false (Section 8).

**6. The caveat extractor.** Earlier: proposers “emit them as structured fields; a two-judge extraction retains only sentences both judges agree carry qualification.” Proposers write prose, and the assembled system uses a document-level prompt that

returns at most one quoted sentence per kind per proposer (Table 4).

**7. The live second-source test.** Earlier: only the rise in the second value was reported. The drop in the first value was not established ( $p = 0.22$ ; Table 5).

**8. Table of main results.** Earlier: intervals for differences were printed beside levels, and cells were shaded by a rule applied to rows without intervals. Table 5 replaces it.

**9. The “source-selector gradient” figure.** Earlier: counts from an early experiment were plotted without the answers that stated neither value (15 to 23 of 27 per arm). The figure is removed.

**10. Judge agreement.** Earlier: “agreement on the deployed corpus is 0.910.” True as raw agreement; chance agreement there is 0.878 and kappa is 0.26.

**11. Counts.** Earlier: “more than fifty registered experiments” and “fourteen instrument-validity findings.” There are 58 registration entries for 61 numbered experiments, and the project’s list of measurement findings now has twenty.

**12. Readings withdrawn or restated.** “Transport of 0.11 to 0.33” as a loss (Section 3.4); “halves corrupted-value adoption”, “is causal” and “0/27 with a clean co-source” (Section 3.3); “self-assessment predicts behavior at chance” and “corroboration is ignored” (Section 3.6); preference shares of 0.818, 0.261 and 0.317 over consistently decided pairs (Section 3.2); “at no cost” (Section 6); the before-and-after figure, which drew separate experiments as one system.

## B. Deviations from Registrations

The project notebook records each of these at the time it happened.

- Experiments 30 and 31: the registered instrument failed its check and was replaced; the registered thresholds were then read as sign tests.
- Experiment 44: the registration did not say over which runs judge agreement was to be computed; the reading that let the result stand was used (0.70 on analysed runs, 0.58 overall; both are reported here).
- Experiment 47: one item’s values were replaced after the pilot, and the pilot check was re-evaluated after the value matcher was repaired.
- Experiment 51: the registration did not say over which items the floor check applied; it was applied to the 11 retained items (0.00) and not to all 18 (0.33).
- Experiment 53: a pilot threshold was lowered from 0.5 to 0.25 after it failed; the experiment then stopped at its checkpoint.
- Experiment 59: the check was re-evaluated after a fourth repair to the value matcher.
- Experiments 60 and 61: token budgets were raised during the experiment after reasoning exhausted them.

- **Timing:** one registration was committed two minutes before its first run and one amendment 33 seconds after; some notebook dates precede the commits that introduced them by up to six days.

### C. Experimental Details

**Phrase-swap (41).** 18 scenarios, 7 repeats, 3 arms, 378 runs. Answers averaged 43.7 to 46.6 sentences across arms. A check refused to run an arm whose prompt contained any registry phrase other than its own.

**Preference (43, 43-R).** 126 pairs per comparison, both orders. Judge 1 gpt-oss:20b; judge 2 a 30B model selected from three candidates on parse rate and consistency only. Pairs decided consistently: 44, 46, 41 (judge 1) and 33, 29, 23 (judge 2). First-listed answer chosen in 0.80 to 0.83 (judge 1) and 0.87 to 0.89 (judge 2) of single judgments. Mean answer lengths 7,757 and 7,649 characters for the pipeline and single answers.

**Second source (47, 59).** 8 items, 5 repeats, 2 arms. Exact match on values after number-format normalisation. In experiment 59 the planning step was told which quantity to cover.

**Caveats in and out (30, 46).** 144 runs, 18 scenarios, two aggregator models, two scenario sets. Supply was varied by removing flagged sentences from proposer text.

**Follow-up (44, 54, 56).** 6 items, 36 runs per arm. Named runs: 7 (informative), 15 (none), 15 (uninformative); 21 of 36 in the live experiment. Fabrication follow-up: 120 replies.

**Appended list (48).** 17 scenarios with at least one flagged sentence, 7 stored answers each, 119 documents, lists of 1 to 53 sentences (median 11). **Reader (51).** 18 items written, 11 retained after the pilot check; 2 readers, 4 arms, 5 repeats, 440 reads.

**Screen (55).** Ten models, two screen prompts, six answers each; 18 pipeline contributions per model.

**Assembled system (61).** 54 runs; 108 pairs against stored plain-pipeline answers; lists of about 5 sentences.

### D. The Sentence Instrument

| Labels from | Kind         | <i>n</i> | raw   | chance | kappa |
|-------------|--------------|----------|-------|--------|-------|
| exp. 41     | assumption   | 15,485   | 0.910 | 0.878  | 0.26  |
| exp. 41     | jurisdiction | 15,485   | 0.833 | 0.717  | 0.41  |
| exp. 41     | hedging      | 15,485   | 0.828 | 0.763  | 0.28  |
| exp. 46     | all three    | 16,869   | 0.838 | 0.752  | 0.35  |
| exp. 30/31  | all three    | 5,697    | 0.833 | 0.772  | 0.27  |

**Table 7:** Agreement between the two sentence judges at the prevalence they are used at. “Chance” is the agreement two judges would reach answering independently at their own base rates.

The instrument was accepted on raw agreement against a bar of 0.70, which independent judges would pass at these base rates. It had been checked on a sample chosen to be rich in positives, where kappa was 0.51 to 0.66, and on twenty anchor sentences written within the project. Of the sentences ei-

ther judge flags, both flag about one in four. On sentences that are positives by construction, both judges flag 0.66 (“modeled at”) and 0.35 (“taken to be”). An earlier version that judged whole documents agreed at 0.62 and was dropped.

Other measurement findings recorded by the project and relevant here: an early judge prompt listed the same seven phrases the writing prompts dictated, found by a registry of every instructed phrase (125 entries); an instruction moved the dictated phrasing from 0/30 to 28/30 answers while other phrasings of the same idea moved from 1/30 to 4/30; and empty judge replies had at one point been scored as negative labels.

### E. Diagnostics That Are Not Optimized

The design records the following and does not tune toward them. *How often a disagreement is named or a follow-up sent:* raising it without new information reproduces the uninformative-reply arm of Section 3.5. *Counts of hedges or instructed phrases:* an instruction raises the phrase (Section 3.1), and the judges mark it down (Section 3.2). *The slope w:* it tracks answer length, and tuning it against the aggregator’s own text would reward caveats the aggregator adds itself. *Judge preference on qualified text:* the judges mostly follow position and mark qualification down. The quantities worth optimizing are checked outside the pipeline’s own text, such as exact-match correctness on items with known answers.